WEEK 1 · DAY 2 · INSTRUCTOR NOTES

Charts That Don't Lie

For the trainer only.

The one idea for today: *the right chart depends on the question.* A breach question needs the worst moment (minimum), not the average; a "who's worst?" question needs small multiples; a "who changed most?" question needs an index. And every chart gets a threshold line and a title that states the finding.

At a glance

Length	~2 hours
Participants leave able to	pick the right daily summary (min vs mean), resample hourly → daily, pivot long → wide, draw small multiples on a shared axis, index to a baseline, choose a y-axis range, write a finding title and a 4-sentence finding
Covers (original)	Week 3 Code Book: **A3 RF link quality** (mean vs minimum) and **B6 VSAT uptime by region** (long format, pivot); Week 3 quiz topic *visualising time-series and data storytelling*
Datasets	`rf_link_quality_hourly.csv`, `vsat_uptime_by_region_weekly.csv` (and `noc_tickets_daily.csv` for one axis example on the slides)
Files you need open	animated slides, your **solution** notebook, the rule book PDF

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Warm-up**	5-minute recap of Day 1: "what makes a time series?" "why `.asfreq()`?" Use Week 3 quiz Q1 and Q6	1–2
0:08	**Hook**	DStv "No Signal" in heavy rain = rain fade. Did our link go down?	3
0:12	**Part 1: the average lies**	Meet the RF data, the hourly chart, **mean vs min** (the killer slide), the rule, cause vs response, one axis	4–9
0:30	**Part 2: six zones**	Long → wide, spaghetti vs small multiples, index to week 1, two questions two answers	10–15
0:45	**Part 3: honest rules**	Y-axis, the 5 rules, title upgrade, 4-sentence finding	16–20
0:55	**Hands-on**	Part A (30 min) then Part B (30 min). Walk the room	21

Time	Block	What happens	Slides
1:50	**Wrap-up**	Scoreboards up; 2–3 volunteers read their 4-sentence findings	22

Explain it like this (plain English)

- **Link margin**: signal left above the minimum, like fuel above the reserve line. Below 0 dB the link is down.
- **The average lies**: August had **12 outage hours on 11 different days**, all during heavy storms. But the *lowest daily average* in the month is **4.91 dB**: comfortably positive. A report on daily averages would say "no outages". The daily **minimum** shows the truth. Rule: *breach questions need the worst moment.*
- **Cause vs response**: rain and uplink power correlate at **0.97**. The station turns up its power *because* it's raining (uplink power control): a response, not a separate fault. Average uplink: **41.2 dBm dry, 44.1 dBm in rain**.
- **One axis**: rain (mm/hr) and power (dBm) are different units. Put them in two stacked charts with a shared time axis, never two y-axes on one chart.
- **Long vs wide**: databases export long (one row per zone per week, 312 rows). Charts want wide (52 rows × 6 columns). `pivot` converts.
- **Small multiples on a shared y-axis**: one mini chart per zone, all with the same scale. If each panel had its own scale, every zone would look equally dramatic.
- **Index to week 1**: `wide / wide.iloc[0] * 100`. Everyone starts at 100; now you see *change from their own normal*.

⚠️ Know this before you teach (the original material is wrong here)

The original code book says *"South South starts lowest **and** falls furthest"*. Only the first half is true:

Question	Answer from the data
Who is lowest?	**South South**: lowest average (97.67%), lowest week 1 (97.99%) and the worst single week (96.56%, mid-September)
Who fell furthest from their own week 1?	**South East**: its indexed line drops to **98.2**. South West reaches 98.3 and South South 98.5

This is actually the best teaching moment of the day: *lowest score ≠ biggest drop since last term.* The southern zones lose about **1 point** of uptime in the rainy season, the northern zones about **0.4**.

Slide talking points

#	Say this	Ask this
3	"Same physics as your DStv in a Lagos storm."	"Who has seen 'No Signal' in heavy rain?"
5	"Only one column matters today: link margin. Fuel above reserve."	—

#	Say this	Ask this
6	Count the dips below the red line with them: 12.	–
7	Show the average first. Pause. "All good?" Then press next: the minimum.	"Which chart would you send your unit head?"
8	Third Mainland Bridge: "average traffic was fine today."	–
9	Correlation 0.97, but uplink power is the *reaction*.	"Does turning up the volume make it rain?"
12	Let them struggle to find South South for 10 seconds.	"Found it?"
13	Point out: same y-axis on all six.	"What would happen with six different axes?"
14–15	Two questions, two answers.	"Which question would your unit head ask?"
17	Uptime zooms in (say the range); tickets start at zero.	"Does Sunday really have no tickets?"
20	Read the 4 sentences slowly. Recommendation last.	–

Answer key

Task	Answer	Common mistake → what to say
A1	`"h"`	`"H"` → newer pandas rejects uppercase; or forgetting quotes
A2	`(rf["link_margin_db"] < 0).sum()` → **12**	`rf["link_margin_db"].min()` gives a value, not a count
A3	`daily_mean.min()` → **4.91**	–
A4	`.min()` → **11** days	leaving `___` → AttributeError mentioning `'___'`
A5	`THRESHOLD = 0`, any title ≥ 20 characters **with a number**	titles without a number fail on purpose
A6	`"min"`	–
A7	`"response"`	–
B1	`"week_start"`, `"region"`, `"uptime_pct"`	swapping index and columns
B2	`True`	`"True"` in quotes also fails: it's a Python value
B3	`wide.mean().idxmin()` → **South South**	typing it by hand with a typo
B4	`indexed.min().idxmin()` → **South East**	expecting South South (see the box above)

Task	Answer	Common mistake → what to say
B5	`"zero"`, `"tight"`	–
B6	Situation, Observation, Implication, Recommendation	–
🌶️	`rf.loc[rf["link_margin_db"] < 0, "rain_rate_mmhr"].min()` → **39.3 mm/hr**	No outage happened below 37.3 mm/hr, and 44 rainy hours had no outage at all

Quiz link

Week 3 quiz (`Time-series/MCQ/week-03-mcq.csv`), topic **"visualising time-series and data storytelling"**, is exactly today: tight y-axis for bounded measures (slide 17), daily minimum for SLA (slide 7), small multiples on a shared axis (slide 13), index to week 1 (slide 14), finding titles (slide 19), recommendation last (slide 20). Use 2–3 of them as Day 3's warm-up.

Troubleshooting

Problem	Fix
`ValueError: Invalid frequency: H`	lowercase `"h"`
`ValueError: Index contains duplicate entries, cannot reshape` in B1	they put `region` as the index
Small multiples squashed or empty	re-run B1's pivot cell; check `wide.shape` is `(52, 6)`
Someone finishes early	Pepper task, then help a neighbour, then draft their 4 sentences

Notes on the original material

- Original code book section *A3* says the RF file is for "mean vs minimum": confirmed. Its daily-mean claim ("stays comfortably positive on every day") is correct: minimum daily mean is 4.91 dB.
- Original section *B6* claims South South "falls furthest": **not true when indexed to week 1** (see above).
- The original uses `.asfreq("H")`, which crashes on pandas 3. Our notebook uses `"h"`.